

Long-Term Health and Sustainability of the PKU Diet: Evidence File

Studies assessing what lifelong low-phe diet plus synthetic protein substitute does to the body and to the people running it. Compiled from the Guthrie PKU corpus (PubMed-indexed; retrieved 2026-09-06).

Bone

[PMID:22295034](#) (Arch Med Sci, 2011) — Two-year DXA follow-up of 32 children and adolescents with classic PKU, designed to test whether the phe-restricted diet is adequate for normal bone mineralization and whether BMD tracks dietary control.

[PMID:12222712](#) (Acta Paediatr, 2002) — In 28 dietary-treated patients aged 10–33, bone density, formation markers, and mineral intake were assessed and osteopenic patients were trialed on low-dose activated vitamin D, despite 78.6% having good dietary compliance.

[PMID:29153023](#) (Orv Hetil, 2017) — Repeat DXA at a 4-year interval in 59 early-treated Hungarian adults, correlating change in bone mineral density with metabolic control; PKU patients have lower BMD than healthy people and the etiology remains unclear.

[PMID:32520722](#) (Endocr Connect, 2020) — Compared 96 diet-treated PKU patients against 62 untreated benign-hyperphenylalaninemia patients on growth, BMI, body composition, and BMD, isolating the treatment's contribution from the disease's.

[PMID:20830525](#) (J Inherit Metab Dis, 2010) — In 47 patients across three age bands, the vegan-like PKU diet's near-total absence of n-3 fatty acids was analyzed as a contributor to bone mass loss via the plasma fatty acid profile's relationship with BMD.

[PMID:30201326](#) (Mol Genet Metab, 2018) — Characterized the osteopenia of the PKU mouse model with histomorphometry to resolve whether hyperphenylalaninemia itself impairs bone differentiation, since human bone-density findings are neither fully penetrant nor uniform.

Micronutrients and B12

[PMID:24113686](#) (Mol Genet Metab, 2013) — Literature review since 1990 of vitamin and mineral deficiency prevalence in dietary-treated PKU, finding inadequacies mainly for zinc and selenium and framing patients as an at-risk group whenever most nutrition comes from non-natural sources.

[PMID:26226119](#) (J Pediatr Endocrinol Metab, 2015) — In 51 long-term-treated patients aged 3–48, vitamin B12 status was assessed with holotranscobalamin, homocysteine, and methylmalonic acid, identifying long-term treatment as a risk factor for B12 deficiency.

[PMID:32482081](#) (Ann Clin Biochem, 2020) — Cross-sectional comparison of 53 diet-treated children and adolescents vs 30 healthy controls using the combined B12 indicator plus methylmalonic acid and homocysteine, on the premise that B12 deficiency frequently appears when the diet is poor in natural protein.

[PMID:29433755](#) (Clin Nutr, 2019) — Case-control study of 112 strictly diet-treated patients without separate vitamin/mineral supplementation vs 36 controls, testing whether the phe-free amino acid formulas alone supply adequate micronutrients.

[PMID:26123187](#) (Mol Genet Metab, 2015) — Multicenter cross-section of 156 hyperphenylalaninemia patients found 81.4% had at least one biochemical marker (selenium, zinc, B12, folate, ferritin, vitamin D, prealbumin)

out of recommended range, though without clinical symptoms.

[PMID:20151202](#) (J Inherit Metab Dis, 2010) — Review framing the phe-restricted diet as nutritionally equivalent to a vegan diet, with the same deficiency concerns, and raising quality questions about the amino acid substitutes themselves.

Body composition, weight, and cardiometabolic risk

[PMID:27896128](#) (Mol Genet Metab Rep, 2014) — BMI data on 947 patients from 8 centres in Europe and Turkey: childhood obesity present at all centres and 5 of 6 adult centres showing elevated female obesity, making excess weight in women the recurring signal.

[PMID:24055312](#) (Mol Genet Metab, 2013) — Review of the overweight question concluding the literature is inconsistent but that reports of increased obesity in female PKU patients keep accumulating, with almost no data on its metabolic consequences.

[PMID:26598928](#) (Ann Nutr Metab, 2016) — Argues obesity, particularly in women, is an emerging PKU comorbidity and proposes minimum monitoring standards (anthropometry, body composition, metabolic comorbidities) as routine clinic practice alongside phe control.

[PMID:38140392](#) (Nutrients, 2023) — Review of cardiovascular risk in PKU tying it to body composition, ectopic fat, high carbohydrate intake, chronic amino acid exposure, and microbiota, and proposing nutritional ultrasound for routine fat-mass surveillance.

[PMID:42212113](#) (Mol Genet Metab Rep, 2026) — Five-year DXA follow-up of 24 early-treated adults: 16 of 24 lost lean mass and gained body fat percentage despite stable weight and BMI, with the changes attributed to the phe-restricted diet.

[PMID:39408322](#) (Nutrients, 2024) — Body composition and biochemical biomarkers in diet-treated adults, analyzed against gender, adherence, and disease severity, motivated by the imbalance created when protein-rich natural foods are excluded and low-phe substitutes included.

[PMID:23137570](#) (Mol Genet Metab, 2012) — The counterpoint: in 89 early-treated patients vs 79 controls with full metabolic-syndrome workups, dietary treatment did not increase risk of obesity or metabolic syndrome.

Growth

[PMID:31484352](#) (Nutrients, 2019) — Systematic review and meta-analysis of long-term growth on the phe-restricted diet: normal at birth and in infancy, but children with PKU were significantly shorter with lower weight-for-age than reference populations.

[PMID:29089407](#) (Pediatrics, 2017) — Retrospective longitudinal analysis of yearly height, weight, and growth-rate measurements in 224 diet-dependent patients aged 0–18 against national database controls.

[PMID:21959353](#) (Clin Nutr, 2012) — Evidence-base review of the repeated finding that early-childhood growth in PKU is sub-optimal relative to controls, and of possible causal links to the dietary management itself.

[PMID:28739202](#) (Mol Genet Metab, 2017) — Two-year prospective study relating protein source (natural vs amino acid formula) and protein-to-energy ratio to growth and body composition, with healthy siblings as controls.

Kidney

[PMID:23138985](#) (J Inherit Metab Dis, 2013) — First systematic renal assessment of the lifelong diet: measured GFR, renal plasma flow, urinary protein, ultrasound, and 24-hour blood pressure in 67 patients aged 15–43, stratified by current diet.

[PMID:37757776](#) (Nephron, 2024) — Ten-year retrospective study of renal function in 41 early-treated adults, premised on amino-acid-mixture-induced glomerular hyperfiltration, comparing kidney outcomes by therapy adherence.

Gut microbiome

[PMID:33806544](#) (Microorganisms, 2021) — 16S sequencing of stool from 11 adults with PKU vs 21 controls found lower gut community diversity in the PKU group, one of the few microbiome datasets in treated adults.

[PMID:27336782](#) (PLoS One, 2016) — Controlled next-generation-sequencing comparison of gut microbiota in 8 diet-treated patients vs 10 healthy individuals, establishing diet-driven compositional differences.

[PMID:41624188](#) (Front Nutr, 2025) — Sibling-controlled pilot isolating the dietary therapy's effect on children's gut microbiota and probing associations with immune function and food allergy.

[PMID:38004705](#) (Microorganisms, 2023) — Mice fed a synthetic free-amino-acid diet modeled on PKU supplements showed decreased bacterial diversity, expansion of disease-associated taxa, reduced colon length, and inflammation over 13 weeks.

Adherence and attrition (sustainability as behavior)

[PMID:23748994](#) (Clin Nurse Spec, 2013) — Qualitative thematic study of adolescents' own accounts of what encourages or inhibits staying on the diet, motivated by persistent apathy and attrition in adolescence.

[PMID:28491816](#) (Mol Genet Metab Rep, 2017) — Longitudinal retrospective study of 75 screened-at-birth children aged 7–13 tracking how adherence erodes through childhood and identifying early warning signs of eventual treatment discontinuation.

[PMID:42370601](#) (J Pediatr Endocrinol Metab, 2026) — Cross-sectional study of adults from newborn screening evaluating which factors predict successful follow-up and treatment continuation after the well-documented adulthood decline in adherence.

[PMID:15956867](#) (J Dev Behav Pediatr, 2005) — Situational analysis interviewing 19 caregivers and 11 children about formula and dietary problems; families who found the regimen less difficult and their coping strategies more effective had children with lower phe.

[PMID:27622144](#) (Mol Genet Metab Rep, 2016) — UK cross-sectional survey quantifying the personal time and out-of-pocket monetary burden that running the low-phe household imposes on caregivers of children with PKU.

Quality of life on and off the diet

[PMID:14971801](#) (Acta Paediatr, 2003) — Fifteen adults who resumed or newly started the restricted diet were interviewed on quality of life; most had poor or only regular dietary control even after resumption.

[PMID:18956249](#) (J Inherit Metab Dis, 2008) — In 53 off-diet adults invited back to the diet, 45% showed moderate or severe psychological distress at baseline, and the study tracked well-being through resumption and analyzed why resumption fails.

[PMID:17160615](#) (J Inherit Metab Dis, 2007) — Course-of-life study of 32 early-and-continuously-treated Dutch young adults, assessing the developmental and psychosocial consequences of growing up on the regimen.

[PMID:21342250](#) (Acta Paediatr, 2011) — Of 41 early-treated patients, only 56% were adherent to prescribed phe intake, with excess intake linked to lower maternal education; the study relates adherence to quality-of-life measures.

[PMID:42255097](#) (Mol Genet Metab Rep, 2026) — The reverse experiment: longitudinal quality-of-life follow-up of classical PKU adults with persistently poor adherence who achieved diet liberalization on pegvaliase, measured before treatment and 6 and 12 months after liberalization.

Synthesis

[PMID:37764724](#) (Nutrients, 2023) — Modified-Delphi expert consensus (13-member international panel, 17 statements agreed) on the long-term effectiveness of medical nutrition therapy, explicitly acknowledging that adherence deterioration from childhood onward complicates any claim about its long-term adequacy.

[PMID:21944883](#) (Mol Genet Metab, 2011) — Adult-outcome review supporting lifelong treatment while conceding that dietary treatment is suboptimal and continuation into adulthood is often not accepted.